

# CHL5233: Quarto in-class assignment

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For this exercise, we will use the following two data sets.

1. `countries.txt`: A data set that contains a list of countries used in the paper “Implications of armed conflict for maternal and child health: A regression analysis of data from 181 countries for 2000–2019” and their three-digit ISO (International Organization for Standardization) code.
2. `regions.txt`: A data set that contains a list of all major countries in the world and their
  - a. ISO code (`country_code`)
  - b. region (`region`)
  - c. sub-region (`sub_region`)
  - d. numerical region code (`region_code`)
  - e. numerical sub-region code (`sub_region_code`)

## In class 20260928

### Read in data and create an analytical data set

| country_name |      | ISO       |      |
|--------------|------|-----------|------|
| Length       | :186 | Length    | :186 |
| N.unique     | :186 | N.unique  | :186 |
| N.blank      | : 0  | N.blank   | : 0  |
| Min.nchar    | : 4  | Min.nchar | : 3  |
| Max.nchar    | : 32 | Max.nchar | : 3  |

```
[1] 186
```

```
[1] 2
```

*There are 186 countries in the countries dataset.*

| country_name  | country_code  | region       | sub_region    |
|---------------|---------------|--------------|---------------|
| Length :249   | Length :249   | Length :249  | Length :249   |
| N.unique :249 | N.unique :249 | N.unique : 6 | N.unique : 18 |
| N.blank : 0   | N.blank : 0   | N.blank : 1  | N.blank : 1   |
| Min.nchar: NA | Min.nchar: 3  | Min.nchar: 0 | Min.nchar: 0  |
| Max.nchar: NA | Max.nchar: 3  | Max.nchar: 8 | Max.nchar: 31 |

| region_code    | sub_region_code |
|----------------|-----------------|
| Min. : 2.00    | Min. : 15.00    |
| 1st Qu.: 9.00  | 1st Qu.: 53.75  |
| Median : 19.00 | Median :154.00  |
| Mean : 65.95   | Mean :179.87    |
| 3rd Qu.:142.00 | 3rd Qu.:202.00  |
| Max. :150.00   | Max. :419.00    |
| NAs :1         | NAs :1          |

## Tabulate countries by region and sub-region

Table 1: Number of countries in each region

| Region   | Count |
|----------|-------|
| Africa   | 54    |
| Americas | 35    |
| Asia     | 46    |
| Europe   | 40    |
| Oceania  | 11    |

Table 2: Number of countries in each subregion

| Subregion                       | Count |
|---------------------------------|-------|
| Australia and New Zealand       | 2     |
| Central Asia                    | 5     |
| Eastern Asia                    | 5     |
| Eastern Europe                  | 10    |
| Latin America and the Caribbean | 33    |
| Melanesia                       | 4     |
| Micronesia                      | 3     |
| Northern Africa                 | 6     |
| Northern America                | 2     |
| Northern Europe                 | 10    |
| Polynesia                       | 2     |
| South-eastern Asia              | 11    |
| Southern Asia                   | 9     |
| Southern Europe                 | 13    |
| Sub-Saharan Africa              | 48    |
| Western Asia                    | 16    |
| Western Europe                  | 7     |

Data visualization using ggplot

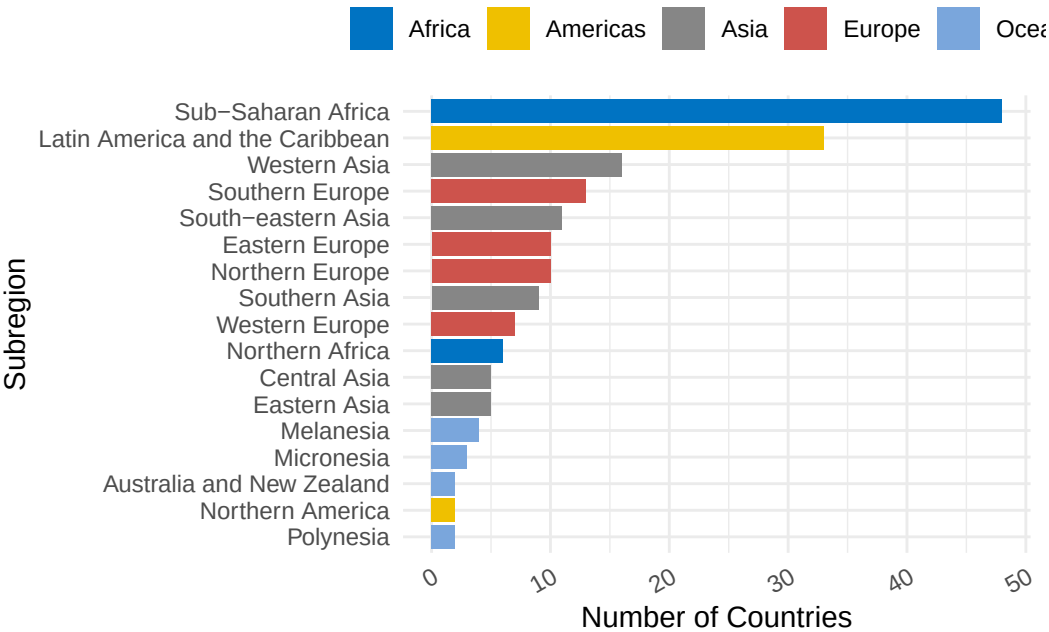


Figure 1: Number of countries per subregion

## In class 20261005

```
# A tibble: 6 x 10
```

|   | iso   | year  | matmor | infmor | neomor | un5mor | Drought | Earthquake | death_sum | conflict |
|---|-------|-------|--------|--------|--------|--------|---------|------------|-----------|----------|
|   | <chr> | <dbl> | <int>  | <dbl>  | <dbl>  | <dbl>  | <dbl>   | <dbl>      | <int>     | <dbl>    |
| 1 | AFG   | 2000  | 1450   | 90.5   | 60.9   | 129.   | 1       | 0          | 5065      | 1        |
| 2 | AFG   | 2001  | 1390   | 87.9   | 59.7   | 125.   | 0       | 1          | 5394      | 1        |
| 3 | AFG   | 2002  | 1300   | 85.3   | 58.5   | 121.   | 0       | 1          | 5553      | 1        |
| 4 | AFG   | 2003  | 1240   | 82.7   | 57.2   | 117.   | 0       | 1          | 1157      | 1        |
| 5 | AFG   | 2004  | 1180   | 80     | 55.9   | 113.   | 0       | 1          | 944       | 1        |
| 6 | AFG   | 2005  | 1140   | 77.3   | 54.6   | 108.   | 0       | 1          | 817       | 1        |